

PROJECT SUBMISSION REPORT

# Crop Doctor

Hybrid AI Crop Health Assistant

Online intelligence. Offline reliability.

|               |                                                                        |
|---------------|------------------------------------------------------------------------|
| Live Website  | boisterous-froyo-7ebad8.netlify.app                                    |
| Category      | Agriculture & Rural Development (AI/ML, Sustainability, Social Impact) |
| Document Type | Detailed Technical Project Report                                      |

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## 01 Abstract

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Crop Doctor is a hybrid AI-powered crop health assistant designed to help farmers identify crop diseases quickly and reliably, regardless of internet availability. The system combines an on-device (offline) machine learning model with a cloud-based (online) analysis service, and automatically routes each request through the best available path. The project consists of a Flutter-based mobile application and a companion web application, both connected to a shared hybrid-AI diagnosis pipeline. The system is designed around the principle of responsible AI: instead of always forcing a diagnosis, it communicates confidence levels, flags disagreement between its online and offline predictions, and recommends expert review when the result is uncertain. This report documents the problem being solved, the proposed solution, the system architecture, the technology stack, the ML model and its evaluation, and the current state of the deployed website and mobile application.

## 02 Introduction

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Agriculture remains the backbone of rural livelihoods in many regions, yet farmers often lack timely access to expert diagnosis when crops show signs of disease or stress. Crop Doctor was built to close this gap using accessible AI technology that works on an ordinary smartphone, with or without a stable internet connection. The project targets the Agriculture & Rural Development theme, with supporting alignment to AI/ML, Sustainability, and Social Impact.

The system is delivered through two connected surfaces: a mobile application built with Flutter, and a web application (deployed and publicly accessible) that mirrors the core diagnosis workflow for demonstration and desktop use.

## 03 Problem Statement

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- Farmers cannot identify crop diseases quickly without expert help, leading to delayed action and visible crop damage before assistance arrives.
- Internet connectivity in rural areas is often weak or expensive, so cloud-only diagnosis tools fail exactly where farmers need them the most.
- Delayed diagnosis translates directly into lost crops and lost income, since many crop diseases spread quickly once symptoms appear.

## 04 Objectives

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- Build a crop disease diagnosis tool that works reliably even without an internet connection.
- Combine an on-device (edge) model with a cloud-based model to improve overall diagnosis quality when connectivity is available.
- Communicate AI results responsibly — showing confidence, flagging uncertainty, and recommending expert review instead of always forcing an answer.
- Deliver the experience through both a mobile application and a web application, using free and open-source tools to keep the solution accessible and low-cost.
- Support local languages so that guidance is understandable to the farmers who need it.

## 05 Proposed Solution

Crop Doctor's core idea is hybrid intelligence: the application does not assume connectivity, it adapts to it. When the farmer captures or uploads a photo of a crop or leaf, the app checks the current internet connection and image quality, then automatically routes the request through the best available path — an on-device TensorFlow Lite model when offline, or an enhanced cloud-based analysis (with live weather context) when online.

|                            |                                                                                                                                                 |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Online Mode</b>         | Cloud AI analysis combined with updated knowledge and live weather data.                                                                        |
| <b>Offline Mode</b>        | On-device AI model runs directly on the phone — no internet required.                                                                           |
| <b>Automatic Switching</b> | The app senses connectivity and image quality, then routes to the best path — the farmer does not need to understand the underlying technology. |

“Crop Doctor doesn't assume connectivity. It adapts to it.” — the farmer gets a useful result whether connected or not.

## 06 System Architecture

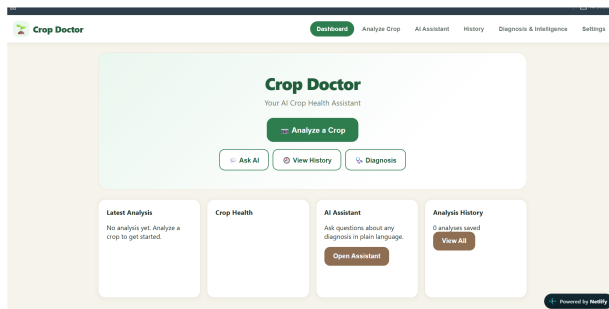
The workflow below traces a single diagnosis request end-to-end, from photo capture on the farmer's device to the guidance ultimately shown to the farmer.

| Step | Stage                                | Description                                                                                                                                                           |
|------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Capture</b>                       | Farmer takes a clear photo of the crop / leaf using the Flutter mobile app (camera or gallery).                                                                       |
| 2    | <b>Route</b>                         | App checks internet connectivity and image quality, then automatically selects the best analysis path.                                                                |
| 3    | <b>Analyze</b>                       | Offline path: on-device MobileNetV3 / EfficientNet-Lite model via TensorFlow Lite. Online path: cloud backend (FastAPI) with enhanced analysis and live weather data. |
| 4    | <b>Compare (Hybrid Intelligence)</b> | Results are compared for agreement or disagreement; uncertainty is handled explicitly rather than forcing an answer.                                                  |
| 5    | <b>Guide</b>                         | Farmer receives the likely issue, confidence level, a simple explanation, recommended action, and an expert-review flag if the result is uncertain.                   |
| 6    | <b>Store</b>                         | Analysis is saved locally (SQLite) and optionally to the cloud (Firebase) for history and monitoring.                                                                 |

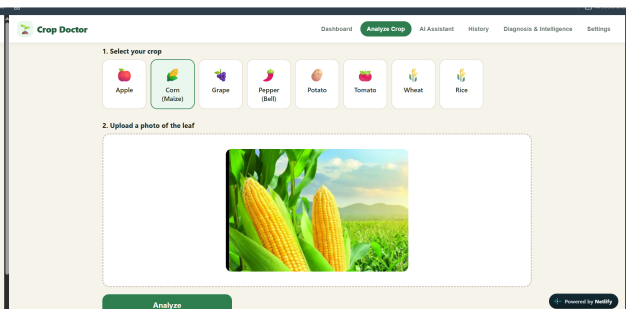
Data & Storage — Local: SQLite (on-device). Cloud: Firebase (optional). This hybrid storage model keeps the app usable offline while still supporting cloud sync when available.

## 07 Website

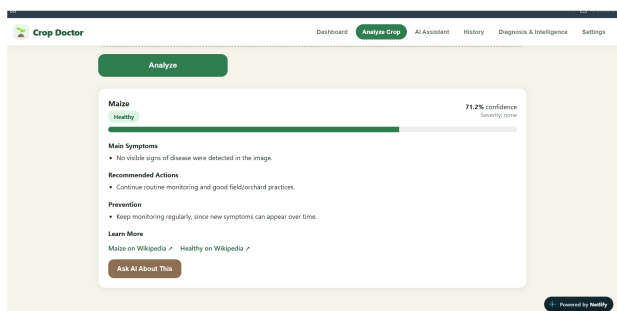
A companion web application mirrors the core diagnosis workflow and is publicly deployed at **boisterous-froyo-7ebad8.netlify.app** (hosted on Netlify). It provides a Dashboard, an Analyze Crop workflow, an AI Assistant chat, Analysis History, and a Diagnosis & Intelligence view.



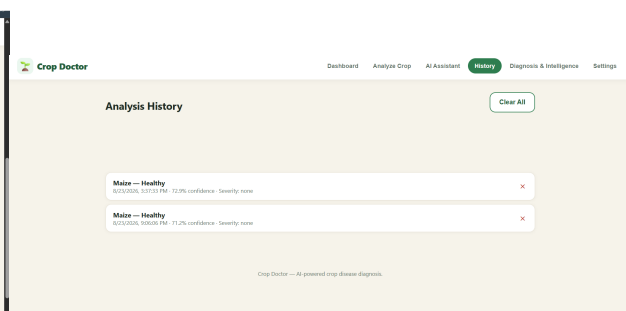
*Dashboard — quick access to Analyze, Ask AI, History and Diagnosis.*



*Analyze Crop — crop selection and leaf photo upload.*



*AI Result — detected condition, confidence, symptoms and recommended actions.*

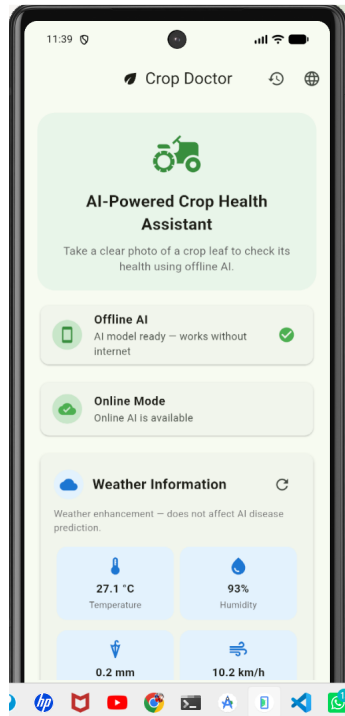


*Analysis History — past scans saved for tracking over time.*

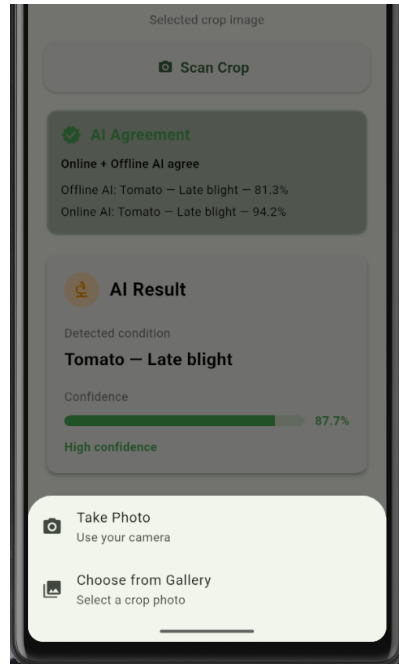
The website also includes an AI Assistant page where users can ask natural-language questions about crop diseases (e.g. “what are common tomato crop diseases?”) and receive plain-language answers, and a Diagnosis & Intelligence page that presents the current diagnosis alongside recommended actions and prevention tips, with links to learn more.

## 08 Mobile Application

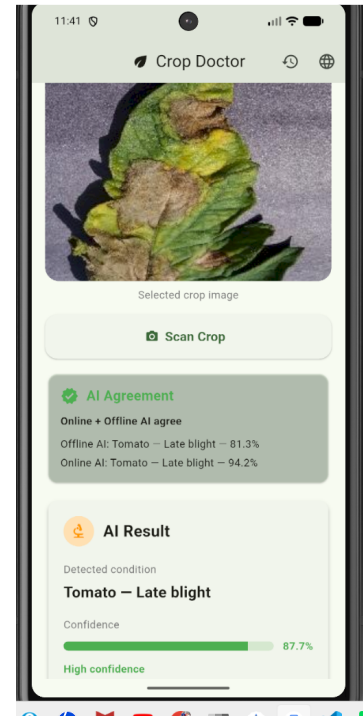
The mobile application is built with Flutter and Dart, chosen for fast, cross-platform development and full offline capability. It mirrors the website's core workflow — capture, analyze, and guide — while being optimized for on-device, low-connectivity use.



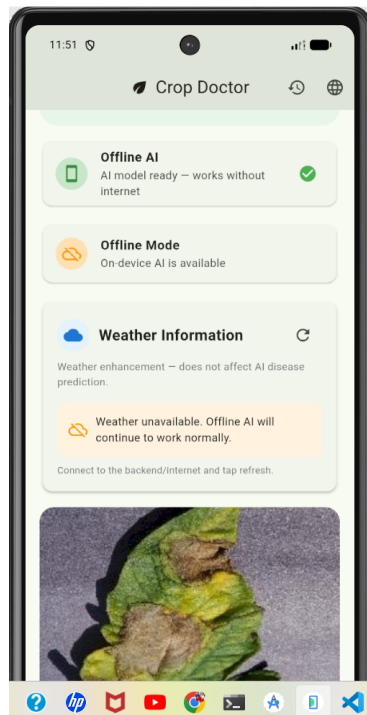
Home — offline/online status, weather info.



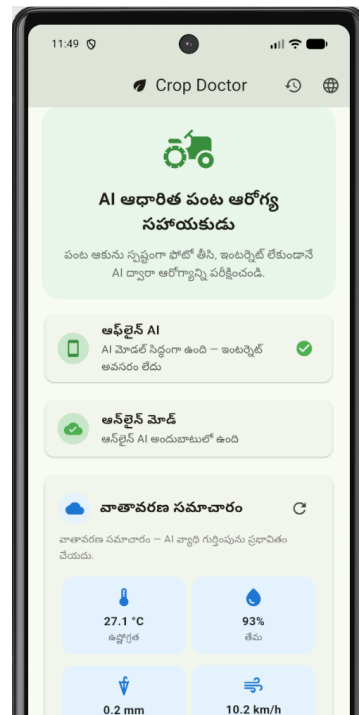
Scan a Crop — camera or gallery capture.



AI Result — condition, confidence, agreement check.



Offline Mode — on-device AI available without internet.



Local Language — Telugu language support.

The app clearly indicates whether Offline AI and Online AI are currently available, and shows live weather information alongside the diagnosis (weather does not affect the AI prediction itself). Language support currently includes English, Hindi, and Telugu.

## 09 ML Model & Dataset

|                           |                                                                                                                  |
|---------------------------|------------------------------------------------------------------------------------------------------------------|
| Offline (Edge) Model      | MobileNetV3 / EfficientNet-Lite, converted to TensorFlow Lite for on-device inference.                           |
| Online (Cloud) Model      | Enhanced analysis served via a FastAPI backend, combined with live weather data.                                 |
| Training Pipeline         | Python with TensorFlow / Keras.                                                                                  |
| Hybrid Intelligence Layer | Compares offline and online predictions for the same image to determine agreement, disagreement, or uncertainty. |

The model architectures (MobileNetV3 and EfficientNet-Lite) were selected specifically for their small footprint and efficiency, making it feasible to run disease classification directly on a farmer's smartphone without depending on a network connection.

## 10 Training & Testing

Both the offline and online models are evaluated on the same input image, and their outputs are compared as a consistency check. Below is a real example observed during testing:

| AI Path    | Result                                 |
|------------|----------------------------------------|
| Offline AI | Tomato — Late blight · Confidence: 87% |
| Online AI  | Tomato — Late blight · Confidence: 94% |

When both AI paths agree, the app treats this as a high-consistency result and delivers guidance with normal confidence. When the two paths disagree, or when confidence is low, the app flags the result as uncertain and recommends expert review rather than presenting a single forced answer.

| Outcome      | System Behaviour                                              |
|--------------|---------------------------------------------------------------|
| Agreement    | High consistency — normal guidance delivered with confidence. |
| Disagreement | Uncertain result — expert review recommended immediately.     |

A separate example captured from the deployed web application: a maize leaf image analyzed as **Healthy** with 71.2%–72.9% confidence and “no visible signs of disease”, with the system recommending continued routine monitoring — illustrating that Crop Doctor reports both disease and healthy outcomes, not just positive disease detections.

## 11 TFLite Integration

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For offline use, the trained model is converted to TensorFlow Lite (TFLite) format and bundled into the Flutter mobile application. This allows disease classification to run directly on the device's processor without any server round-trip, which is what enables the app's core diagnostic function to keep working when the farmer has no internet access at all. The TFLite model output (predicted class and confidence) is combined with the on-device Hybrid Intelligence layer to decide whether the result should be shown directly or flagged for review.

## 12 Features

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- Hybrid Online + Offline AI diagnosis, with automatic routing based on connectivity.
- Confidence and uncertainty handling — honest results rather than forced answers.
- Multi-language support — English, Hindi, and Telugu in the mobile app.
- AI Assistant chat for follow-up questions in plain language, on both web and mobile.
- Weather-aware context shown alongside diagnosis results.
- Analysis History to track and revisit past scans over time.
- Diagnosis & Intelligence view with recommended actions, prevention tips, and learn-more links.

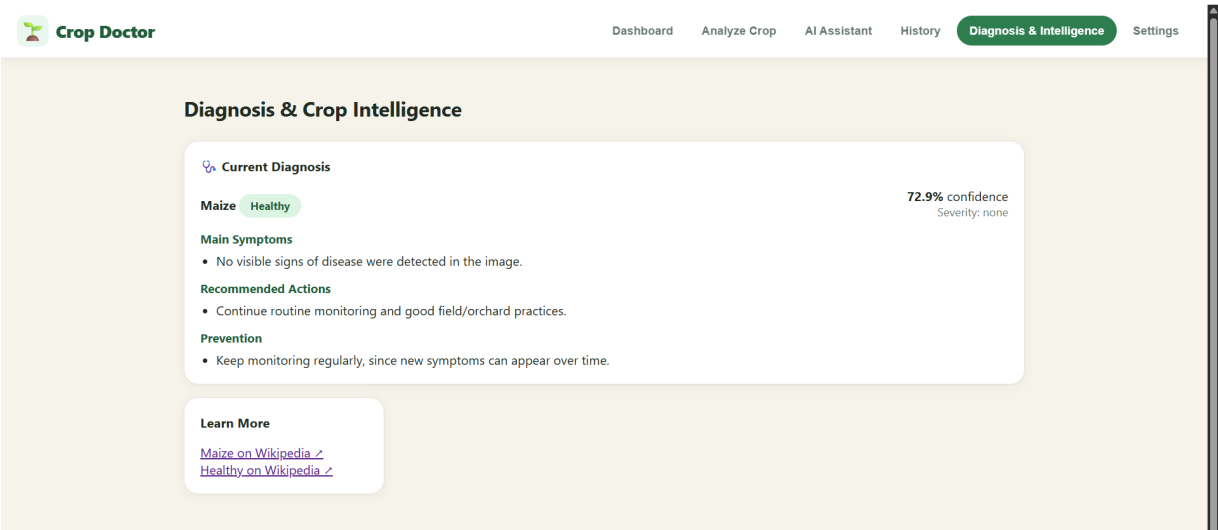
## 13 Results

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The system has been exercised end-to-end on the deployed website and the mobile application, using real crop images captured through the app and browser. Observed behaviour from testing includes:

- Correct routing between offline and online analysis paths based on live connectivity and image quality checks.
- Consistent agreement between offline and online predictions on the same image (example: Tomato — Late blight at 87% offline vs. 94% online confidence).
- Accurate reporting of healthy crops as well as diseased ones (example: Maize — Healthy at 71–73% confidence).
- Analysis History correctly logging past scans with crop, condition, timestamp, confidence, and severity.
- Functional AI Assistant chat answering natural-language questions about crop diseases (e.g. tomato disease symptoms) in plain language.





Diagnosis & Intelligence view with recommended actions and prevention tips.

14 Security

- Images stay on the phone by default — there are no forced cloud uploads; cloud analysis is used only for the online path when the user has connectivity.
- Local-first storage (SQLite) keeps analysis history on-device; cloud sync via Firebase is optional.
- Responsible AI messaging: the app explicitly avoids presenting a diagnosis as certain when confidence is low or the two AI paths disagree, reducing the risk of farmers acting on an unreliable result.

15 Scalability

Crop Doctor is deliberately scoped as an MVP with a clear expansion path — the current build covers 2–3 crops (rice, wheat, cotton) and selected disease classes, demonstrated on Android with English and Telugu support. The team's stated position is that this is not a one-crop demo, but the first stage of a platform.

| Stage | Focus                   | Description                                             |
|-------|-------------------------|---------------------------------------------------------|
| 01    | Field Validation        | Test the MVP with real farmers and gather feedback.     |
| 02    | Crop Expansion          | Grow coverage from 2–3 crops to 10+ crops.              |
| 03    | Regional Models         | Introduce climate- and geography-aware crop models.     |
| 04    | Multi-language Platform | Extend local-language support across more regions.      |
| 05    | Expert Network          | Integrate agricultural expert review into the workflow. |

The system is built entirely on free and open-source tools with no dependency on expensive sensors or special hardware — only a smartphone (and, for development, a laptop) — which keeps the cost of scaling low.

## 16 Real-World Impact

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- Faster access to crop-health information in low-connectivity areas, where cloud-only tools would otherwise be unusable.
- A knowledge bridge that translates technical plant-pathology information into simple, local-language explanations farmers can act on.
- Responsible AI behaviour — confidence, uncertainty, and expert escalation — that builds trust rather than encouraging blind reliance on a single automated answer.
- Hybrid intelligence that lets the same tool serve both connected and disconnected farmers without maintaining two separate products.

## 17 Limitations

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- The current MVP covers a limited set of crops (2–3) and selected disease classes rather than the full range of crops and diseases farmers may encounter.
- The Android demo and English/Telugu language support have not yet been extended to all platforms and languages the project aims to support.
- Predictions are preliminary guidance, not a medical-grade diagnosis — the system is explicitly designed to direct farmers to qualified agricultural professionals rather than replace them.
- Field validation with real farmers, at scale, is still an upcoming step rather than a completed one.

## 18 Future Scope

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- Expand crop and disease coverage from 2–3 crops toward 10+ crops.
- Introduce regional, climate-aware models tailored to different growing conditions.
- Broaden multi-language support across additional local languages and regions.
- Integrate a network of agricultural experts directly into the review workflow for uncertain cases.
- Continue field validation with real farmers to refine accuracy and usability based on direct feedback.

## 19 Conclusion

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Crop Doctor demonstrates that a hybrid online/offline AI architecture can make crop disease diagnosis accessible to farmers regardless of internet connectivity, using only free and open-source tools and an ordinary smartphone. By combining an on-device TensorFlow Lite model with a cloud-based FastAPI service, and by handling uncertainty responsibly instead of forcing an answer, the project delivers a technically feasible, low-cost, and scalable foundation for real-world agricultural support. The current MVP — covering a small set of crops with Android and web deployments — is intentionally scoped as a starting point for field validation and expansion into a broader crop-health platform.

## 20 References

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- Live deployed website: [boisterous-froyo-7ebad8.netlify.app](https://boisterous-froyo-7ebad8.netlify.app)
- TensorFlow Lite — on-device machine learning inference framework.
- MobileNetV3 / EfficientNet-Lite — lightweight convolutional neural network architectures for edge deployment.
- FastAPI — Python web framework used for the cloud analysis backend.
- Flutter & Dart — cross-platform mobile application framework.
- Firebase — optional cloud storage and sync service.
- Netlify — website hosting and deployment platform.